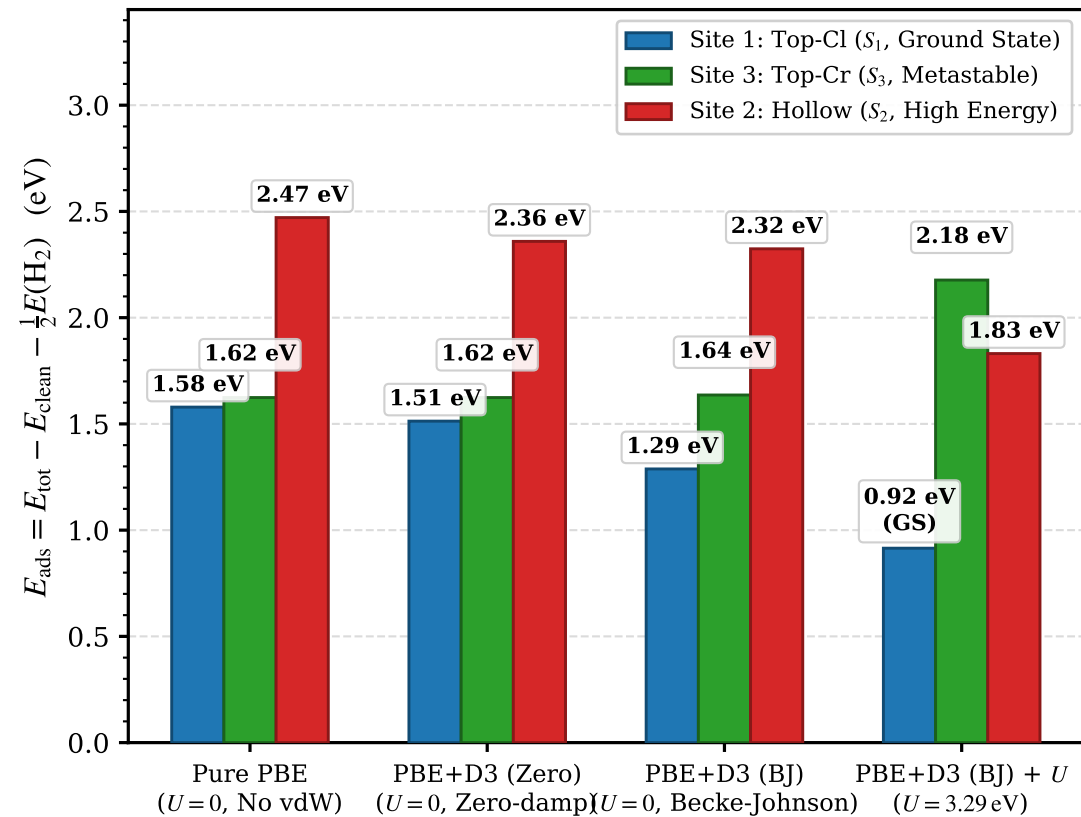
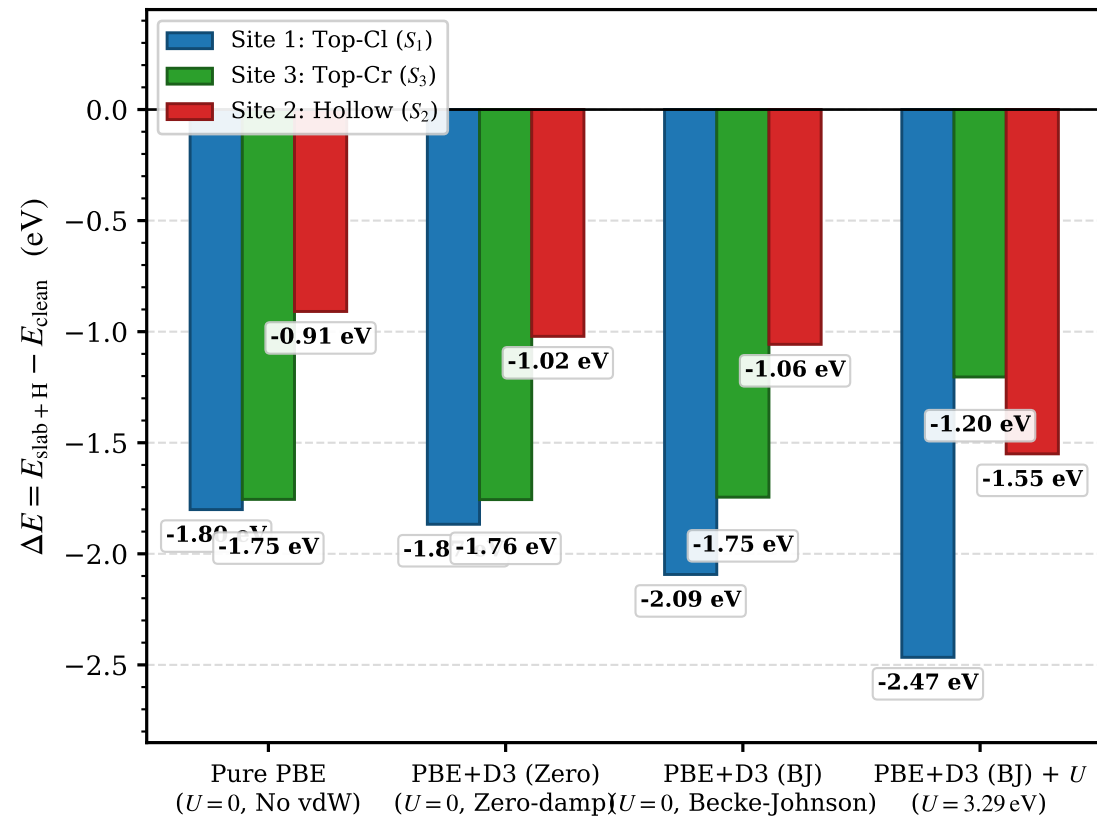


# Comprehensive Hydrogen Adsorption Benchmark on $2 \times 2$ Monolayer $\text{CrCl}_3$ ( $\theta = 0.25$ , $d_{\text{H-H}} = 12.09 \text{ \AA}$ )

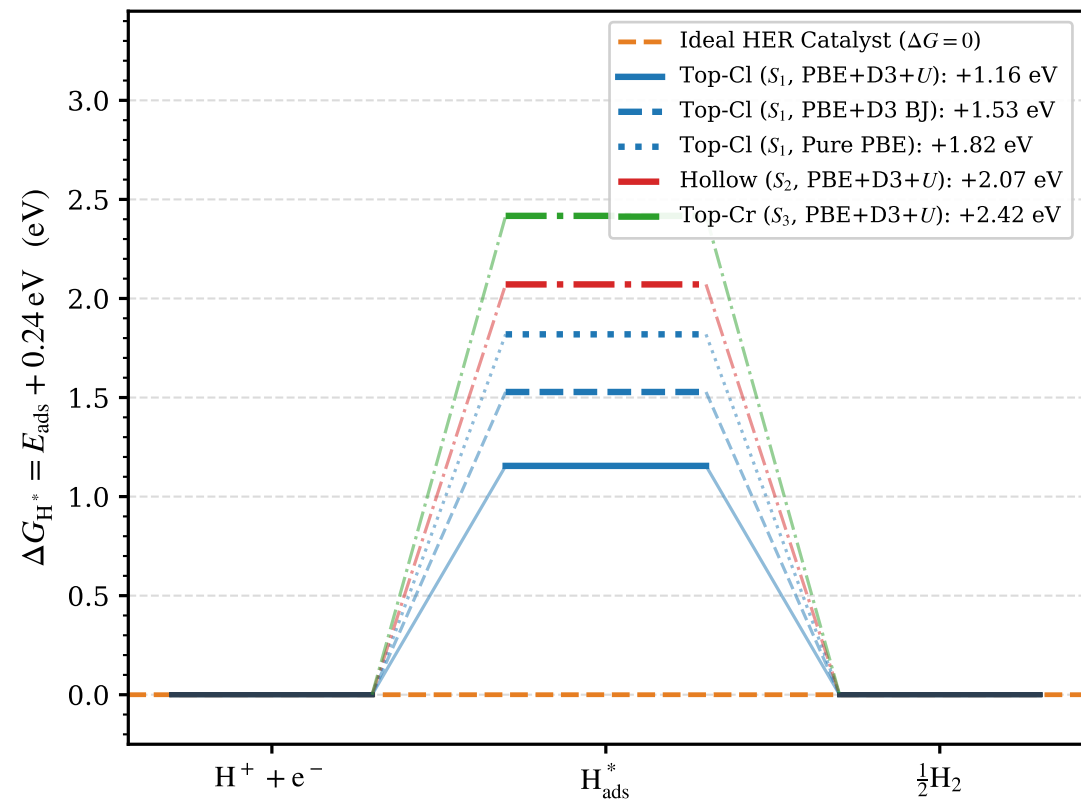
**(a)  $2 \times 2$  Hydrogen Adsorption Energy ( $E_{\text{ads}}$ )**



**(b) Thermodynamic Binding Energy ( $\Delta E$ )**



**(c)  $2 \times 2$  HER Free Energy Profile ( $\theta = 0.25$ )**



**(d) Spin Polarization & Magnetic Compensation ( $2 \times 2$ )**

